

LEGAL SAHARA

Software Requirement Specification (SRS)

https://github.com/rafsharahim1/fyp2025-2026_Legal_System

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Software Requirement Specification (SRS)

1. Problem Statement

1.1 Research and Case Discovery Crisis

Pakistani lawyers face a "research and case discovery crisis" characterized by:

- *Manual search inefficiency*: 3-5 hours per case for precedent research
- *Publication delays*: Printed law reports lag 1-2 years behind actual decisions
- *Inadequate digital tools*: Existing databases offer only basic keyword matching, incapable of understanding semantic context or factual similarity
- *Geographic disparities*: Lawyers in smaller cities lack access to comprehensive legal libraries

1.2 Documentation and Drafting Inefficiencies

Current document preparation challenges include:

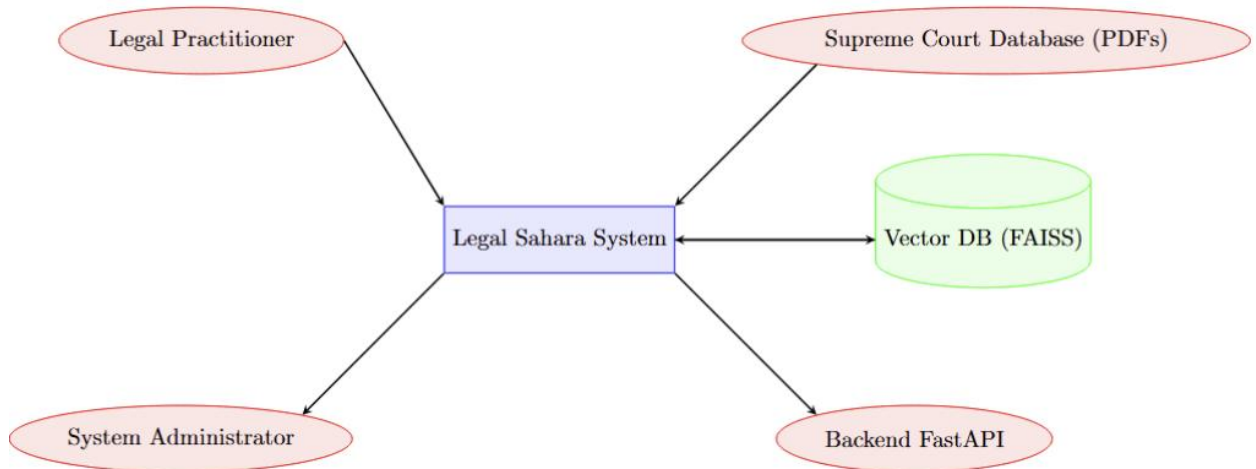
- *Time-intensive drafting*: 3-5 hours to draft a single bail petition
- *High error risk*: Accidental citation of overruled or invalidated precedents
- *Lack of standardization*: No intelligent drafting tools to generate first drafts based on case facts
- *Inconsistent quality*: Manual processes lead to variations in document quality and legal argumentation

1.3 Systemic Impact

These inefficiencies create:

- A digital divide where access to quality legal support depends on firm size rather than case merits
- Reduced court efficiency due to poorly researched petitions
- Barriers for junior lawyers without proper research support
- A two-tiered justice system favoring well-resourced urban firms

Legal Sahara Solution: Functions as a "hyper-competent junior associate" that understands natural language, performs deep semantic research, and autonomously drafts accurate legal documents in minutes.



2. System Requirement

2.1. List of Functional Requirements

ID	REQUIREMENT	DESCRIPTION
FR-1	USER AUTHENTICATION	USERS MUST BE ABLE TO CREATE AN ACCOUNT, LOG IN, AND LOG OUT SECURELY.
FR-2	SEMANTIC CASE LAW SEARCH	USERS MUST BE ABLE TO SEARCH THE SUPREME COURT DATABASE USING NATURAL LANGUAGE QUERIES AND FACT PATTERNS, NOT JUST KEYWORDS. (THE RAG COMPONENT).
FR-3	AGENTIC DOCUMENT DRAFTING	THE SYSTEM SHALL PROVIDE AN AGENT THAT CAN AUTONOMOUSLY DRAFT LEGAL PETITIONS (E.G., BAIL, WRIT) BASED ON A USER'S PROMPT.
FR-4	ENTITY PARSING (NER)	THE AGENT MUST AUTOMATICALLY PARSE THE USER'S PROMPT TO EXTRACT KEY LEGAL ENTITIES (CLIENT NAME, PPC SECTION, FIR NO.) USING A LEGAL NER MODEL (E.G., OPENNYAIORG).
FR-5	VALIDATION & FOLLOW-UP	THE AGENT MUST INTERNALLY VALIDATE IF ALL REQUIRED INFORMATION IS PRESENT (E.G., FIR_NO FOR BAIL) AND ASK THE USER FOLLOW-UP QUESTIONS IF IT'S NOT.
FR-6	PRECEDENT RETRIEVAL	THE DRAFTING AGENT MUST AUTONOMOUSLY CALL THE SEMANTIC_SEARCH_TOOL TO FIND AND CITE RELEVANT CASE LAW IN THE GENERATED DRAFT.
FR-7	CITATION NETWORK ANALYSIS (RESEARCH)	(RESEARCH COMPONENT) THE SYSTEM SHALL MODEL THE CASE LAW AS A CITATION GRAPH AND USE ALGORITHMS (E.G., PAGERANK) TO PROVIDE AN "AUTHORITY SCORE" FOR PRECEDENTS, ENHANCING SEARCH.
FR-8	DOCUMENT EXPORT	USERS MUST BE ABLE TO DOWNLOAD THE FINAL GENERATED DRAFT AS A PROFESSIONAL, COMPILABLE .TEX (LATEX) FILE.
FR-9	AUTOMATED CASE SUMMARIZATION	THE SYSTEM SHALL PROVIDE A TOOL (E.G., CASE_SUMMARIZATION_TOOL) THAT THE AGENT CAN USE TO GENERATE CONCISE SUMMARIES OF RETRIEVED LEGAL JUDGMENTS.
FR-10	DYNAMIC TEMPLATE REPOSITORY	THE SYSTEM SHALL MAINTAIN A REPOSITORY OF LEGAL DOCUMENT TEMPLATES (E.G., BAIL_PETITION.TXT,

FR-11	SPECIALIZED LEGAL LLM	WRIT_PETITION.TXT) THAT THE AGENT CAN AUTONOMOUSLY LOAD.
		(RESEARCH COMPONENT) THE AGENT'S CORE GENERATIVE AND REASONING "BRAIN" MUST BE A DOMAIN-SPECIFIC LLM (E.G., SAULLM-7B-INSTRUCT-V1) TO ENSURE HIGH-QUALITY, ACCURATE, AND NON-HALLUCINATORY LEGAL DRAFTING.

2.2. List of Non-functional Requirements

ID	REQUIREMENT	DESCRIPTION
NFR-1	PERFORMANCE (SEARCH)	SEMANTIC SEARCH QUERIES SHOULD RETURN RESULTS IN UNDER 12 SECONDS.
NFR-2	PERFORMANCE (DRAFTING)	A FULL PETITION DRAFT (INCLUDING PARSING, VALIDATION, SEARCH, AND GENERATION) SHOULD BE COMPLETED IN UNDER 2 MINUTES.
NFR-3	ACCURACY (SEARCH)	THE SEMANTIC SEARCH SHALL ACHIEVE A TARGET RELEVANCE SCORE OF 85%.
NFR-4	ACCURACY (GENERATION)	THE LLM-GENERATED DRAFT MUST BE FACTUALLY CONSISTENT WITH THE USER'S PROMPT AND NOT HALLUCINATE NEW, INCORRECT CASE LAW.
NFR-5	USABILITY	THE USER INTERFACE MUST BE CLEAN, SIMPLE, AND INTUITIVE, CENTERED AROUND A SINGLE PROMPT/CHAT INTERFACE
NFR-6	DATA INTEGRITY	THE CASE LAW DATABASE (VECTOR INDEX) MUST BE BUILT FROM VERIFIED SUPREME COURT JUDGMENTS.
NFR-7	SCALABILITY	THE SYSTEM ARCHITECTURE (FASTAPI BACKEND, VECTOR DB) MUST SUPPORT ALL 977+ INITIAL PDFS AND SCALE TO INCLUDE FUTURE HIGH COURT JUDGMENTS.
NFR-8	MODULARITY	THE SYSTEM MUST BE BUILT AS A SET OF INDEPENDENT "TOOLS" (PARSER, SEARCHER, TEMPLATE LOADER) ORCHESTRATED BY A CENTRAL AGENT (LANGCHAIN) TO ALLOW FOR EASY UPGRADES.

2.3. User Interface Requirements

The user interface will follow a single-screen, chat-centered design, ensuring minimal complexity and maximum usability. The primary interaction model is a chat-style workflow, where all communication between the user and the agent occurs within a continuous conversational stream. The central element of the UI is the Chat Display Area, which presents system messages, extracted entities, follow-up questions, previews of retrieved case summaries, and final drafted outputs. Below this is a wide Input Bar, which supports multi-line text for long factual descriptions and includes basic validation to prevent empty submissions. To maintain transparency during processing, the interface incorporates Status Indicators that show real-time system actions such as "Extracting legal entities...", "Validating required fields...", "Retrieving relevant case law...", and "Generating LaTeX draft...". Once the draft is complete, the Document Output View displays a confirmation message along with a clearly visible button to download the generated .tex file. An optional Context Sidebar may also be included, primarily to support supervisor evaluation and debugging; this panel can display extracted entities, summarized case law used in generation, and citation graph details such as Authority Scores, enabling users to understand how the system arrived at its conclusions.

Parsed Entities

Contract Law

Breach of Duty

Negligence

Citation Score

Low

High

Find precedents related to breach of contract in software development projects where the scope of work was poorly defined.



Certainly. I've found three landmark cases that address breach of contract due to ambiguous scope of work. Here are the summaries:

Innovate Corp. v. Solutions Inc.

★ High Authority

This case established that if the scope of work is fundamentally ambiguous, the burden of clarification falls on the service provider. The court ruled in favor of the client, setting a key precedent for "duty to inquire."

View Full Case

Save to Library

TechLeap v. DataWeave

★ Medium Authority

A pivotal case where continuous "agile-like" verbal agreements were not sufficient to override an initially ambiguous written contract. The ruling highlighted the importance of written change orders.

View Full Case

Save to Library

Ask about a legal precedent or case...



Retrieved Case Summaries

Innovate Corp. v. Solutions Inc.

...established that if the scope of work is fundamentally ambiguous, the burden of clarification falls on the service provider.

TechLeap v. DataWeave

...where continuous "agile-like" verbal agreements were not sufficient to override an initially ambiguous written contract.

QuantumBuild v. FinSoft LLC

This appellate court decision focused on the "reasonable person" standard in interpreting vague SOWs in the tech industry.

Draft a New Petition

Describe the key facts of your case below. Our AI will analyze the information and generate a preliminary petition draft in LaTeX format.

Enter case facts, plaintiff/defendant details, desired relief, etc.

 Generate ✨

Generating draft...

Retrieving case law...

Generated Petition (LaTeX)

 Copy  Download .tex

```
1 \documentclass{article}
2 \usepackage{geometry}
3 \geometry{a4paper, margin=1in}
4 \title{Petition}
5 \author{John Doe}
6 \date{\today}
7
8 \begin{document}
9
10 \maketitle
11
12 \section{Introduction}
13 This is a sample petition generated by Legal Sahara. The plaintiff
14
15 \section{Parties}
16 \begin{enumerate}
17 \item The Plaintiff, John Doe, is an individual residing at 123 Main St, Anytown, CA 90210.
18 \item The Defendant, Jane Smith, is an individual residing at 456 Elm St, Anytown, CA 90210.
19 \end{enumerate}
20
21 \section{Facts}
22 On or about January 1, 2023, the following events occurred...
23
24 \section{Prayer for Relief}
25 WHEREFORE, Plaintiff prays for judgment against Defendant for:
26 \begin{itemize}
27 \item General damages in an amount to be determined at trial.
28 \item Special damages for medical expenses and lost wages.
29 \item Punitive damages.
30 \item Costs of suit.
31 \item Such other and further relief as the Court may deem just.
32 \end{itemize}
33
```

Administration Panel

Home / Admin Panel

System Controls

Core System Functions

Manage high-level system configurations and AI models.



 Re-index Vector Database

 Manage AI Models

Template Management

 Add New Template

Bail Application

Criminal Law

Writ Petition

Constitutional Law

Non-Disclosure Agreement

Corporate Law

Recent Activity Logs

 Filter logs by user or action...

USER	ACTION	TIMESTAMP
admin@legalsahara.com	Logged In	2023-10-27 10:30 AM
support@legalsahara.com	Deleted Template: "Divorce Petition"	2023-10-27 09:15 AM
admin@legalsahara.com	Updated Template: "Bail Application"	2023-10-26 04:55 PM
jane.doe@clientfirm.com	Generated document from template "NDA"	2023-10-26 03:20 PM

3. Functional Requirements Specification

Provide use cases based on 1 and 2 above.

3.1. Stake Holders

- **Lawyers** – primary users
- **Junior staff & Paralegals** – drafting + research
- **Law students** – learning tool

3.2. Actors and Goals

Actor 1: Legal Practitioner

Goals:

1. Retrieve high-quality precedents.
2. Draft accurate petitions with minimal manual editing.

3. Identify missing details.
4. Export the final document professionally.

Actor 2: System Administrator

Goals:

1. Manage legal templates.
2. Re-index vector database when new cases are added.

3.3. Use Cases

3.3.1 Use Case List

- *UC-001*: User Login
- *UC-002*: Semantic Search
- *UC-003*: Draft Petition (Agentic Workflow)
- *UC-004*: Missing Info Handling
- *UC-005*: Case summarization

3.3.2 UC-003 (Draft Legal Petition)

Use Case ID: UC-003

Trigger: User enters prompt: “Draft a bail petition for ...”

Actor: Legal Practitioner

Preconditions

1. User is logged in (FR-1).
2. NER model and LLM loaded (FR-4, FR-11).
3. Tools initialized (parser, search, loader).
4. Vector DB indexed.

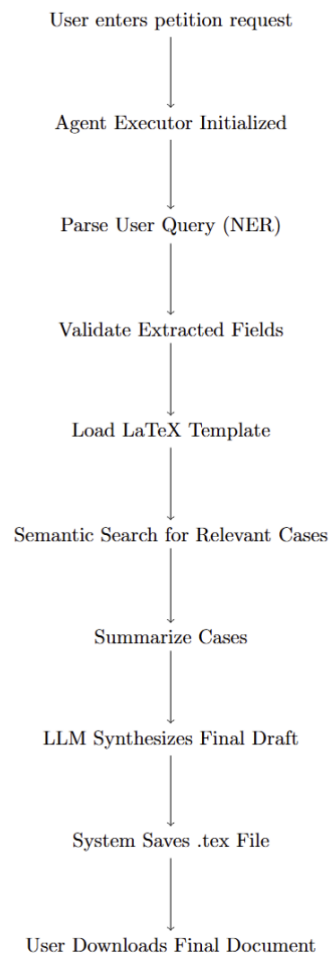
Main Success Scenario:

1. User enters request into prompt interface
2. System invokes **LangChain Agent Executor**
3. **Agent Thought**: "I must understand this request"
4. **Agent Action**: Calls `parse_user_query_tool(prompt=...)` (FR-4)
5. **Tool Observation**: Returns JSON: `{"petition_type": "bail", "client_name": "Zoya Akhtar", "fir_no": "123/2025", ...}`
6. **Agent Thought**: "Validate parsed data"
7. **Agent Action**: Runs `validate_and_get_follow_ups` logic (FR-5)
8. **Agent Observation**: All required fields present—validation successful
9. **Agent Thought**: "Need template"
10. **Agent Action**: Calls `load_document_template_tool(template_filename="bail_petition.txt")` (FR-10)
11. **Tool Observation**: Returns full LaTeX template string
12. **Agent Thought**: "Need case law"
13. **Agent Action**: Calls `semantic_search_tool(query="complainant filed FIR after delay...")` (FR-2, FR-6)

14. **Tool Observation:** Returns JSON of case summaries (enhanced by FR-9)
15. **Agent Thought:** "Synthesize final draft from all components"
16. **Agent Action:** Generates final LaTeX text using SaulLM-7B (FR-3)
17. System saves output as .tex file (e.g., Zoya_Akhtar_bail.tex)
18. System returns "Status: Complete" message with downloadable file path (FR-8)

Exception Flow:

- Agent detects missing fields.
- Agent asks targeted questions.
- User provides missing info.
- Workflow resumes from validation step.



3.3.2 UC-002 — Semantic Case Law Search (RAG Workflow)

Use Case ID: UC-002

Trigger: User enters a natural-language legal query into the search bar

Actor: Legal Practitioner

Preconditions

1. User is logged in (FR-1).
2. Vector database is fully indexed with embeddings (FR-2).
3. Semantic search tool is initialized and connected to FAISS (FR-6).
4. Summarization tool is loaded (FR-9).
5. Citation Graph + Authority Score model (optional) is loaded (FR-7).

Main Success Scenario

1. User enters query

User types:

“Show me cases where FIR was delayed and the accused was granted bail.”

2. System invokes LangChain Agent Executor

The agent takes control of the multi-tool workflow.

3. Agent Thought:

“I must interpret this search request.”

4. Agent Action:

Calls *parse_user_query_tool(prompt=...)* (FR-4).

5. Tool Observation:

Returns structured JSON:

```
{  
  "query_type": "semantic_search",  
  "legal_issue": "delayed FIR",  
  "context": "bail jurisprudence",  
  "keywords": ["FIR delay", "bail allowed", "SC precedent"]  
}
```

6. Agent Thought

“Now embed the query to perform semantic retrieval.”

7. Agent Action:

Calls *embed_query_tool(text="delayed FIR bail cases")*

8. Tool Observation:

Returns eg 768-dim embedding vector.

9. Agent Thought:

“Retrieve most similar case chunks from vector DB.”

10. Agent Action:

Calls *semantic_search_tool(embedding_vector=..., top_k=10)* (FR-2, FR-6)

11. Tool Observation:

FAISS returns top-K case chunks with metadata:

- case_name
- PLD/SCMR reference
- similarity score
- paragraph text

12. Agent Thought (optional, if enabled):

“Re-rank results using Authority Score (PageRank).”

13. Agent Action:

Calls *citation_graph_rerank_tool(results=...)* (FR-7)

14. Tool Observation:

Returns list re-ordered by precedent authority.

15. Agent Thought:

“Summarize retrieved cases to reduce cognitive load for user.

16. Agent Action:

For each retrieved case, calls:

case_summarization_tool(case_text=...) (FR-9)

17. Tool Observation:

Returns concise summaries (150–250 words).

18. Agent Thought:

“Compile final response with ranked cases + summaries.”

19. System Output:

User sees a structured search result panel:

Ranked by Relevance / Authority Score

- Case 1: Summary + Score
 - Case 2: Summary + Score
 - Case 3: Summary + Score
 - ...
- (Up to K results)

20. System returns status message:

“Semantic Search Complete — 10 relevant precedents found.”

Post-Conditions

- Search log saved to database (optional logging).
- Retrieved cases ready for use in drafting workflows (FR-3).

Exception Flow

Case 1 — Query Too Short

Agent detects insufficient info

Asks user:

“Please describe the factual background for better search accuracy.”

Case 2 — No Relevant Cases Found

Agent responds:

“No directly matching cases found. Would you like:

→ broader search?

→ search only High Court cases?

→ keyword-based search?”

Case 3 — Vector DB Not Indexed

System returns warning:

“Search unavailable. Please contact administrator.”

Case 4 — Summarizer Failure

System displays:

“Raw case text returned (summary unavailable).”

4. User Interface Specification

UC-002: Search for Case Law

- User enters a legal/factual query.
- System returns:
 - Ranked cases
 - Low-latency summaries
 - Authority Score (if enabled)

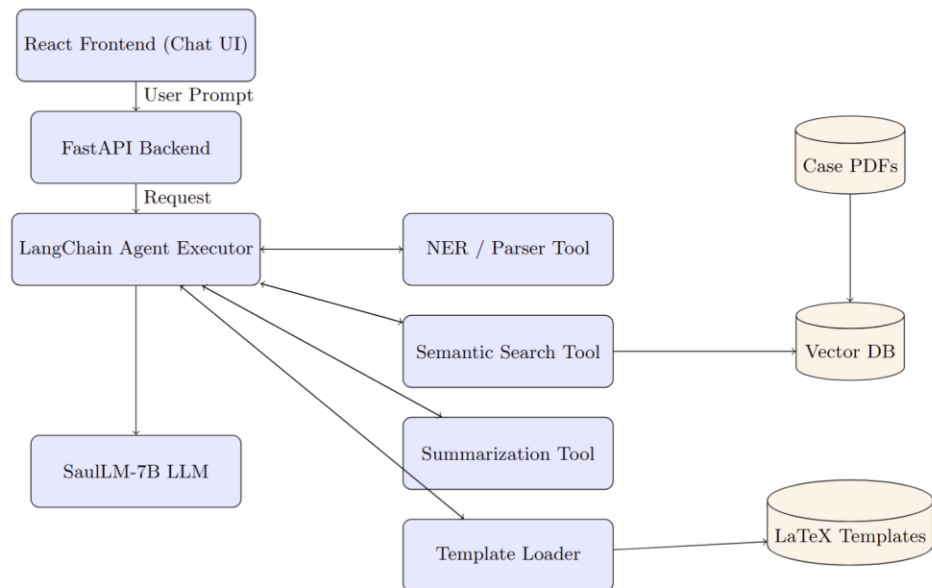
UC-003: Draft Petition

- User provides case facts.
- Agent displays intermediate steps (optional):
 - Parsed entities
 - Retrieved cases
- Final output is a LaTeX file.

UC-004: Missing Info

- If fir_no is missing → ask directly.
- Reply integrated into the ongoing workflow, not starting over.

5. Domain Analysis

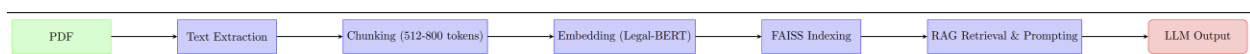


5.1 Model 1: Retrieval-Augmented Generation (RAG)

Purpose: Retrieve factually relevant precedent text to ground LLM outputs.

Detailed RAG Pipeline

1. *Parsing PDFs*
 - Convert scanned PDFs → text
 - Remove headers, footers, artifacts
2. *Chunking*
 - Split into 512–800 token chunks
 - Maintain citation metadata
3. *Embedding*
 - Legal-BERT → 768-D vectors
4. *Indexing*
 - FAISS: HNSW or FlatIP
5. *Query → Retrieval*
 - User query embedded
 - K-NN search returns top-k chunks
6. *Augmented Prompting*
 - Retrieved chunks injected into LLM prompt
7. *LLM Generation*
 - Ensures no hallucinated cases (NFR-4)



5.2 Model 2: Citation Network Graph (Research Component) (Optional)

Domain Model: Model Supreme Court judgments as *Directed Graph* $G = (V, E)$:

- V (*Vertices*): All judgments (e.g., PLD 2022 SC 112)
- E (*Edges*): Citations where edge (A, B) means "Judgment A cites Judgment B"

Mathematical Model (PageRank):

Calculate Authority Score (AS) for every judgment:

$$AS(v_i) = \frac{1 - d}{N} + d \sum_{v_j \in M(v_i)} \frac{AS(v_j)}{L(v_j)}$$

Where:

- d = damping factor
- N = total number of cases
- $M(v_i)$ = cases citing v_i
- $L(v_j)$ = outbound links from v_j

Novel Hybrid Retrieval Algorithm:

Combine RAG and Graph models:

$$FinalScore(D) = (\alpha \times SemanticScore(D)) + ((1 - \alpha) \times AuthorityScore(D))$$

Where α is a tunable weighting factor. This ensures relevant results are ranked higher if they are also authoritative "VIP" cases.

5.3 Model 3: Fine-Tuning Strategy for Pakistani Legal Context

To optimize Legal Sahara for Pakistani legal language, terminology, and procedural nuances, we implement a comprehensive fine-tuning strategy across multiple model components.

5.3.1 Legal-BERT Fine-Tuning for Embeddings

Objective: Adapt the pre-trained Legal-BERT model to better understand Pakistani legal terminology, Urdu-English code-mixing, and domain-specific concepts.

Fine-Tuning Approach:

1. *Dataset Preparation:*
 - Curate corpus of 977+ Pakistani Supreme Court judgments

- Extract text segments containing legal terminology, case citations, and procedural language
- Create parallel corpus of semantically similar case pairs for contrastive learning

5.3.2 Named Entity Recognition (NER) Fine-Tuning (Optional)

Objective: Customize the NER model to accurately identify Pakistan-specific legal entities.

Fine-Tuning Approach:

1. *Custom Entity Types:*
 - CLIENT_NAME: Petitioner/respondent names
 - PPC_SECTION: Pakistan Penal Code sections (e.g., "302", "376")
 - FIR_NUMBER: First Information Report numbers
 - COURT_NAME: Pakistani court identifiers (e.g., "Supreme Court of Pakistan")
 - CASE_CITATION: Pakistani case citations (e.g., "PLD 2022 SC 112")
 - DATE: Arrest dates, hearing dates, judgment dates
 - POLICE_STATION: Station names and jurisdictions
2. *Training Data Annotation:*
 - Manually annotate 500-1000 Pakistani legal documents
 - Use semi-automated annotation tools with human verification
 - Create balanced dataset across different entity types
3. *Evaluation Metrics:*
 - Precision, Recall, F1-score per entity type
 - Target: >90% F1-score for critical entities (PPC_SECTION, FIR_NUMBER)

Conclusion:

The Legal Sahara system represents a significant step toward modernizing Pakistan's legal research and drafting ecosystem by integrating Retrieval-Augmented Generation, semantic search, citation network analysis, and domain-specific language models into a unified intelligent platform. Through its modular tool-based architecture, the system addresses the critical pain points identified in the problem domain—slow research cycles, inconsistent drafting quality, lack of semantic retrieval, and limited access to authoritative judgments. By combining advanced AI models with verified legal datasets and a transparent, chat-based user interface, Legal Sahara ensures accuracy, efficiency, and reliability in both precedent discovery and petition drafting. The solution not only enhances productivity for legal practitioners but also reduces systemic inequities by providing junior lawyers and resource-constrained firms with capabilities comparable to well-resourced institutions.

